5-MINUTE REVISION

Probability

From the chapter's own revision points, formulae and questions.

Selected from Chapter 10 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- Random experiment → experiment whose outcome cannot be predicted with certainty.
- Sample space → set of all possible outcomes.
- Sample point → one possible outcome.
- Random variable → numerical value assigned to outcomes.
- Discrete random variable → takes countable values.

Quick revision · 2

- Continuous random variable → takes values within a range.
- Probability → measure of the likelihood of an event.
- Probability always lies between 0 and 1.
- Impossible event → probability 0.
- Certain event → probability 1.

Quick revision · 3

- Event → one or more outcomes of a random experiment.
- Independent events → occurrence of one does not influence the other.
- Dependent events → occurrence of one influences the other.
- Mutually exclusive events → cannot occur together.
- Exhaustive events → together cover the entire sample space.

Quick revision · 4

- Equally likely events → have equal probability.
- Classical probability → based on exhaustive, mutually exclusive, and equally likely outcomes.
- Statistical probability → based on repeated observations.
- Axiomatic probability → based on the three axioms of probability.
- A.N. Kolmogorov → associated with the axiomatic approach.

Quick revision · 5

- Complement → event not occurring.
- $A \cup B$ → A or B or both.
- $A \cap B$ → both A and B.
- A^c → complement of A.
- Conditional probability → probability of an event given that another event has occurred.

Quick revision · 6

- Multiplication law → used for probability of joint occurrence.
- With replacement and independent random selection → successive draws have the same probabilities and are independent.
- Without replacement → successive selections from a finite set are generally dependent.
- Combination → selection without considering order.
- Bayes' theorem → obtains the probability of a cause or hypothesis given observed evidence.

Quick revision · 7

- Prior probability → probability before considering the evidence.
- Posterior probability → updated probability after considering the evidence.
- Gambler's fallacy → belief that previous independent outcomes influence future outcomes.
- In independent trials, past outcomes do not change the probability of the next outcome.

Important formulae · 1

Probability of an event under equally likely outcomes:

$$P(A) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

Classical probability:

$$P(E) = \frac{m}{n}$$

where m is the number of favourable outcomes and n is the total number of exhaustive, mutually exclusive, and equally likely outcomes.

Statistical probability:

$$P(E) = \lim_{n \to \infty} \frac{m}{n}$$

Important formulae · 2

Probability range:

$$0 \leq P(E) \leq 1$$

Probability of sample space:

$$P(S) = 1$$

Complement rule:

$$P(A^c) = 1 - P(A)$$

Important formulae · 3

Probability of an event and its complement:

$$P(A) + P(A^c) = 1$$

Additive law of probability:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Additive law for mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

Important formulae · 4

Conditional probability:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Multiplication law:

$$P(A \cap B) = P(A)P(B \mid A)$$

Multiplication law for independent events:

$$P(A \cap B) = P(A)P(B)$$

Important formulae · 5

Combination:

$$^nC_r = \frac{n!}{r!(n-r)!}$$

Bayes' theorem for two events:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}, \quad P(B) \neq 0$$

Bayes' theorem for a partition of the sample space:

$$P(E_i \mid A) = \frac{P(E_i)P(A \mid E_i)}{\sum_{k=1}^{n} P(E_k)P(A \mid E_k)}$$

Worked example from the chapter

A fair coin has been tossed 99 times, with heads on every toss. Assuming independent tosses, what is the probability of heads on the 100th toss?

Solution

Answer is $\frac{1}{2}$ as each toss is independent of previous toss.

Self-check from the chapter

1. The set of all possible outcomes of a random experiment is called the _____ space. **[Sample]**
2. A subset of the sample space is called an _____. **[Event]**
3. A real-valued function of the outcomes of a random experiment is called a random _____. **[Variable]**
4. Events that cannot occur together are called mutually _____ events. **[Exclusive]**